COL707 · ICTs FOR DEVELOPMENT AND SUSTAINABILITY
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Classic Readings for ICTD Project Teams

Governance, appropriate technology, equity, participation, and justice — foundational lenses for scoping and evaluating an ICTD project

1. Ostrom — Governing the Commons
2. Schumacher — Small Is Beautiful
3. Unwin — Technology for the Power-Weak
4. Freire — Pedagogy of the Oppressed
5. Rawls — A Theory of Justice
6. Young — Justice and the Politics of Difference
7. Sen — Development as Freedom
8. Seth — Technology and (Dis)Empowerment

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ROADMAP

Four lenses, one question

How should communities, technologies, and technologists relate to one another — and what does that mean for the project you design?

I

Governing shared resources

Elinor Ostrom

Can communities govern their own commons — without the state or the market?

II

Appropriate technology

E. F. Schumacher

What makes a technology usable, ownable, and manageable by the people who depend on it?

III

Building for the power-weak

Tim Unwin

Who is technology actually designed to benefit — and who gets left further behind?

IV

Working with, not for

Paulo Freire

How does change happen with communities rather than being delivered to them?

I

Governing Shared Resources

Elinor Ostrom — Governing the Commons (1990)

Neither the state nor the market has solved it

The tragedy of the commons assumes a shared resource is doomed unless it is centrally controlled by government or privatized into individual ownership.

State control

Centralised rules, imposed from outside. Often fails to fit local conditions.

Privatisation

Splits the resource into individual plots. Loses the benefits of shared stewardship.

Ostrom's question

Can the users of a resource design and enforce their own rules — and sustain that self-governance over time?

What Ostrom set out to explain

1. What increases the initial likelihood that a group will self-organise at all?
2. What enhances a group's capacity to keep self-governing over time?
3. What lets a group solve its common-pool-resource problems without relying on external assistance?

Complexities to handle along the way: opportunistic appropriation, understanding the resource's own structure, free-riding, trial-and-error learning, discounting future benefits, and accountability for non-compliance.

Ostrom's design principles for durable commons

Institutions she found among long-enduring, self-governed common-pool resource systems

1 Clearly defined boundaries

Who has rights to withdraw from the resource, and the boundaries of the resource itself, are clearly known.

2 Congruence with local conditions

Rules for use and provision fit local needs and conditions, not a generic template.

3 Collective-choice arrangements

Those affected by the rules can participate in modifying them.

4 Monitoring

Monitors — often accountable to the users, or the users themselves — audit resource conditions and behaviour.

5 Graduated sanctions

Rule-breakers face sanctions that scale with the seriousness and context of the offence.

6 Conflict-resolution mechanisms

Low-cost, accessible venues exist for resolving disputes between users or with officials.

7 Minimal recognition of rights to organise

External authorities do not challenge the community's right to devise its own institutions.

8 Nested enterprises

For larger commons, governance, monitoring and sanctioning are organised in nested layers.

Rules in practice: Village - Kali, Panchayat - Dumaro, Block - Chandawa, Dist – Latehar. See rules noted in these field trip report.
https://docs.google.com/document/d/10UUlOW8K-7dSg8-_X34l6bQh8uz8FVnC/edit?usp=sharing&ouid=105881081359587580379&rtpof=true&sd=true

Monitoring and sanctions have to move together

Monitoring

Produces private benefits for the monitor and joint benefits for everyone else — e.g. “ditch riders” who regularly patrol to check for misuse.

Graduated sanctions

Punishment scaled to the offence — first infractions treated lightly, repeated ones more seriously — keeps commitment credible without alienating members.

Why both, together: rule-following stays high only when infractions are visible and consequences are credible. Remove either piece, and higher rates of rule-breaking feed further rule-breaking — a collapse of the institution. Together, they let individuals make a contingent self-commitment: “I will follow the rule if I trust that others are being watched and held to it too.”

Digital commons need governance, not just openness

Making data, a community network, or a knowledge repository “open” does not make it self-sustaining. Ostrom's principles translate directly:

Boundaries	Who can contribute to / withdraw from a shared dataset, model, or channel — clearly defined membership.
Local fit	Rules for a community radio archive or farmer data-pool should match local norms, not a platform default. For example, see this paper.
Monitoring & sanctions	Someone has to check for misuse of shared data or infrastructure, with consequences that scale.
Nested governance	A federation of local knowledge commons, rather than one centrally administered platform.

II

Appropriate Technology

E. F. Schumacher — Small Is Beautiful (1973)

Two kinds of mechanisation

Enhances human skill and power

A tool that extends what a person can already do — puts capability and judgement in the hands of the user.

Turns the worker into a mechanical slave

A system so complex or centralised that the person must instead serve it — leaving them “in a position of having to serve a slave.”

The question worth asking of any technology is whether it liberates or subordinates the person using it.

Intermediate technology — the middle path

More productive than inefficient traditional tools, but less costly and more manageable than large-scale, capital-intensive industrial technology.

1. Creates workplaces where people are already living, not where they are forced to migrate
2. Cheap enough that workplaces can be created in large numbers
3. Simple enough that skill demands stay low — in production, organisation, supply, financing, and marketing alike
4. Uses local materials, mainly for local use

The intellectual infrastructure behind appropriate technology

Communication

Lets each field worker know what other work is going on nearby, geographically or functionally.

Information brokerage

Enables effective sharing of what has already been learned, so it isn't relearned from scratch.

Feedback

Carries technical problems from the field back to places with the skills to solve them.

Action groups & verification centres

Coordinate sub-structures that create and test solutions locally.

The digital equivalent: own and manage, even without building end-to-end

Communities rarely write the code, design the chips, or run the cloud behind a digital tool. Schumacher's argument still applies at the layer that matters most: whether people can understand, adapt, and manage what they depend on.

Not required

Building every layer locally — hardware, core software, cloud infrastructure — from scratch

What matters instead

Enough local understanding to configure, repair, adapt content, question the design, and decide whether to keep using it — ownership of use, not just of code

What this means for a project design

Local capacity building	Train community operators, not just end-users — mirrors Schumacher's field-worker network.
Repairable, adaptable design	Favour tools that local technicians can fix and extend over black-box systems.
Ownership of process	Community decides whether and how to keep using the tool — not locked into a vendor's roadmap.
Right-sized scale	Match the technology's complexity to the skills and resources actually available on the ground.

III

Building for the Power-Weak

Tim Unwin — Why the Notion of a Fourth Industrial Revolution is So Problematic (2019)

Technology doesn't change the world — people do

Every “industrial revolution” narrative rests on the assumption that technology itself is the changing force. Unwin argues the opposite:

> *“Technology is designed by people for particular purposes, that serve very specific interests, and it is these that change the world, not technology.”*

Humans still have a choice: design technologies that make the rich and powerful richer and more powerful, or craft technologies that empower and serve the poor and marginalised.

A self-fulfilling prophecy

The interests shaping these technologies are not primarily about making society fairer or more equal — they are about sustaining dominance and wealth.

> *“The Fourth Industrial Revolution is in large part a conspiracy to shape the world ever more closely in the imagination of a small, rich, male and powerful élite.”*
>
> — Tim Unwin, UNESCO Chair in ICT4D

Whose story gets told?

Unwin calls out an “elite view of history” built into how technological change is usually narrated:

Owners of factories	*not the labourers who work in them*
Innovative geniuses	*not the peasants labouring to produce food for them, or public resources consumed to build the innovations without clearly conceptualizing how the innovations can lead to public good*
Wise politicians	*not the communities whose lives their policies reshape*
Male heroes of the revolution	*not the women whose roles are written out*

The imperative: build for the poorest, or the gap only widens

“The only route forward is to work towards empowering the poorest and most marginalised people to create a world that is better in their eyes than in our own.”

— Tim Unwin, 2019

Without this deliberate choice, technology defaults to serving those who already hold resources, skills, and power — and the inequality gap widens rather than closes.

A design and evaluation checklist question

Who does this project make richer or more powerful — and who does it leave further behind?

1. Who chose the problem this project solves — and who wasn't in the room?
2. Does the benefit flow mainly to intermediaries (platforms, vendors, officials) or to the intended users?
3. If this tool succeeds exactly as designed, does the gap between the powerful and the power-weak shrink?

IV

Working With, Not For

Paulo Freire — Pedagogy of the Oppressed (1968)

Two logics that keep change from happening

The oppressor's logic

Any restriction on their way of life feels like a violation of individual rights — with no matching concern for those who suffer from hunger and despair. If others don't have more, it must be because they are incompetent, lazy, or ungrateful for generosity already shown.

The oppressed's false consciousness

The peasant feels inferior to the boss because the boss seems to be the only one who knows how to run things — rarely realising he too knows things learned through his own relations with the world and with other people.

Critical consciousness: the turning point

The oppressed realise that the oppressor cannot exist without them. The peasant gains the courage to overcome dependence the moment he recognises that he is dependent.

How this comes about — not through slogans

It needs authentic, humanist generosity, not paternalism — “a real humanist can be identified more by his trust in the people, which engages him in their struggle, than by a thousand actions that favour without that trust.” Propaganda and monologue, however well-intentioned, only turn people back into masses that can be manipulated.

Dialogue, not deposits of knowledge

The wrong model

Treating people as empty “buckets” to be filled with the teacher's knowledge — a one-way transfer that leaves existing understanding unrecognised.

Freire's model

The teacher presents material for consideration; the students' own considerations reshape it; this evokes new understanding on both sides. People are “buckets” already full — dialogue is how thinking is shaped together.

“Communion with the people” in practice

Freire's discussion of cooperation includes this account from Che Guevara, of his time as a doctor among peasants in the Sierra Maestra:

“As a result of daily contact with these people and their problems we became firmly convinced of the need for a complete change in the life of our people. **Communion with the people ceased to be a mere theory, to become an integral part of ourselves.**

Guerrillas and peasants began to merge into a solid mass. No one can say exactly when, in this long process, the ideas became reality and we became a part of the peasantry... those poor, suffering, loyal inhabitants of the Sierra cannot even imagine what a great contribution they made to the forging of our revolutionary ideology.”

Praxis: reflection and action, together

Action without reflection

→ Activism. Motion without direction or learning.

Reflection without action

→ Verbalism. Talk that never changes the situation.

Reflection + Action

→ Praxis. The source of knowledge and genuine transformation.

Praxis re-presents the world not as a settled thought, but as a problem — one that guerrillas (and technologists) alike only understand by being inside it, not observing from outside it.

Cultural synthesis: reframe the demand as a shared problem

Narrow aspiration

“We want a salary increase” — a demand framed and settled entirely within the existing structure.

Posed as a problem

“Are we sellers of our labour, or owners of it? Any sale of labour is a form of slavery” — leaders pose the deeper question, and people organise around the synthesis that follows.

For ICTD teams: the way a need is framed with a community — as a request for a gadget, or as a question about who controls the process — shapes what kind of change becomes possible.

V

Design With People, Politically

Participatory design and action research — from Freire’s dialogue to political practice

Design is political, not just collaborative

Two threads in the PD literature — where the movement came from, and why participation on its own is not enough

Born in the labour movement — Gregory, 2003

PD began in Scandinavian workplaces when computers arrived for time-keeping and electronic publishing — often as tools of managerial control, not worker benefit. Landmark projects such as the Norwegian Iron and Metalworkers' project and the Scandinavian UTOPIA project with typographers treated workers' tacit shop-floor knowledge as real expertise, with trade unions as partners, not subjects. Conflict was embraced, not smoothed over — power should be visualised, not hidden. The method itself was praxis: muddle along, gain experience, and learn by doing, close to Freire's own reflection joined to action.

Participation is not enough — Beck, 2002

Beck's ‘P is for Political’ warns that a process can look fully participatory — workshops, prototyping, sign-off — while the underlying power relations stay untouched. PD must be motivated by political conscience at every stage of development and adoption, not only at the point of consultation. The deeper question for a project is not ‘did we consult the community?’ but ‘who shapes the agenda — the research, design, and development agenda itself?’

For ICTD teams: ask not only whether the community participated, but whether the process was designed to shift power — and who controlled the agenda before your team arrived.

Learning by being part of the action

Hayes, 2011: The relationship of action research to HCI — and its kinship with Freire's method

Praxis, not observation
Action research asks researchers to work alongside participants, cycling through plan, act, and reflect — not to observe from a distance for 'objective' findings. That spiral mirrors Freire's own praxis: reflection and action together, understood only from inside a situation, never from outside it.

Not value-neutral
AR requires researchers to state their relation to the people and the work up front — the same demand Freire made of anyone claiming solidarity with the oppressed: a commitment to improving the situation, not a project managed and handed down from the top.

Local solutions, not universal proof
Success is judged by whether a workable, transferable solution emerged for this situation, not by generalisability — just as Freire's method could never be reduced to a fixed technique detached from a particular people's reality.

For ICTD teams: your fieldwork is itself a form of praxis — plan, act, and reflect alongside the community, not only about it.

Learning by being part of the action

Hayes, 2011: The relationship of action research to HCI — and its kinship with Freire's method

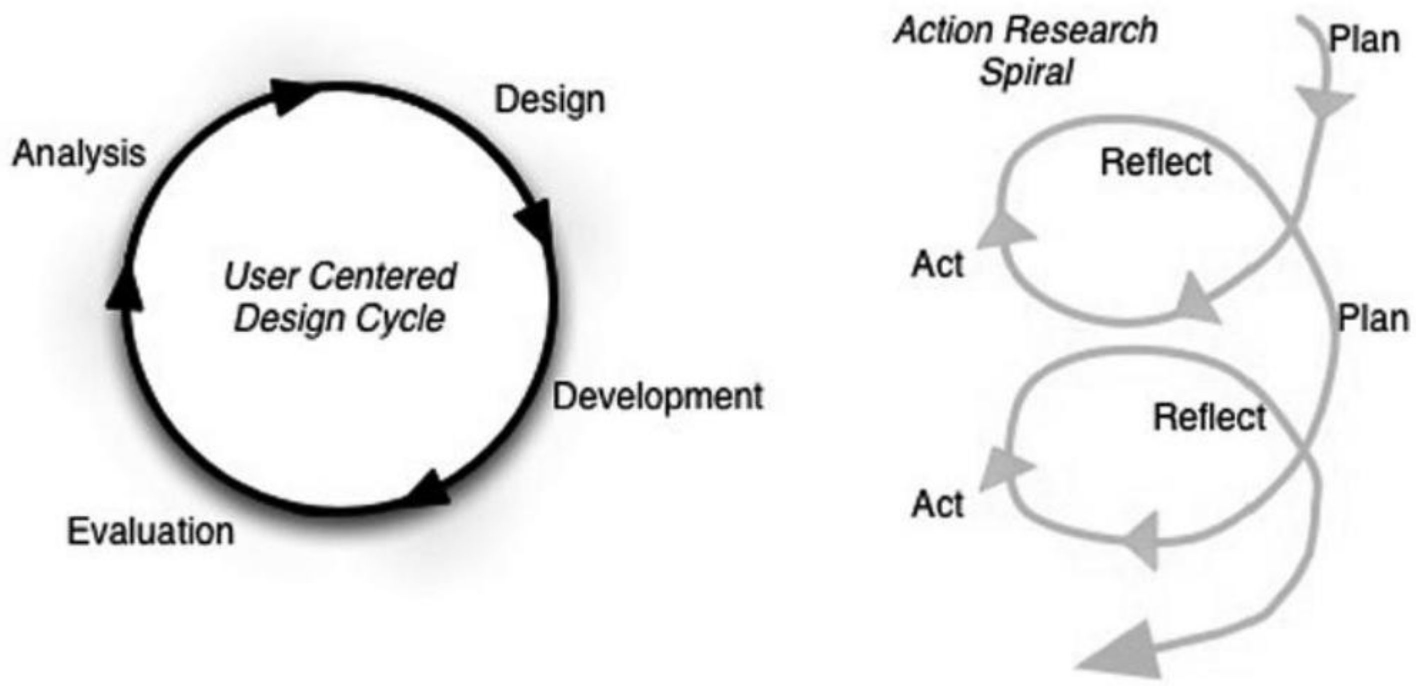


Fig. 1. User-Centered Design is commonly depicted as a cyclical process of design, development, evaluation, and analysis. AR is more commonly represented as a spiral, as in this figure, but is also sometimes represented as a cycle of planning, action, and reflection.

Four lenses, one habit of mind

Ostrom	Can this community govern its own shared resource — with clear rules it enforces itself?
Schumacher	Can the people who depend on this technology understand, adapt, and manage it?
Unwin	Does this project make the powerful more powerful, or does it close the gap for the power-weak?
Freire	Is this change being done with the community, through dialogue — or delivered to it?

VI

Development, Justice, and Democracy

Rawls, Young, and Sen on what justice requires — and why Seth insists democracy is what makes it possible

The veil of ignorance and two principles of justice

A Theory of Justice, 1971 — how could people ever agree on a fair social contract?

The veil of ignorance

Imagine deciding the rules of society before you know your own place in it — your sex, race, ethnicity, wealth, intellectual capability, or physical abilities are all hidden behind a veil of ignorance. Rawls argues that rational people reasoning from behind this veil, unable to design rules that favour themselves, will converge on principles that are genuinely fair to whoever they might turn out to be — rich or poor, majority or minority.

Two principles of justice

First: each person may claim a fully adequate set of basic rights and liberties — thought, speech, association, safety from harm, ownership of property — so long as this is consistent with everyone else having the same claim.

Second, the difference principle: any social and economic inequalities must attach to positions everyone has fair and equal opportunity to reach, and must work to the greatest benefit of society's least-advantaged members.

Left unanswered: what exactly are the 'basic liberties' or the 'inequalities' being measured — only material goods? And who gets to decide? Young and Sen press on both.

Justice is about structure, not just distribution

Justice and the Politics of Difference, 1990 — a top-up to Rawls’ model of social welfare

Beyond distributive justice

Rawls' theory is a theory of distributive justice — the difference principle only asks whether material goods end up fairly shared. It says nothing about the unjust social processes that produced the inequality in the first place: the oppression and domination of specific social groups — Black people, gay people, the elderly, women — through norms that a distribution-only lens simply cannot see, let alone change.

Recognise difference, then politicise it

Countering that oppression requires recognising group-based difference explicitly, and politicising it within a democratic setup — moving away from the veil of ignorance rather than reasoning from behind it. Like Sen, Young's end-state is freedom from oppression and domination: the fairness of the prevailing social norm or reality, not just the fairness of the resulting distribution. And justice should not only judge a situation against an ideal — it should show a path to reach it.

Democracy, for Young, is the only setup that allows the ongoing reflection and change this requires.

Development as the freedom to achieve

Development as Freedom, 1999 — equality of what, exactly?

What Rawls leaves open

Rawls never specifies what end-state of equality is envisioned — equality of what? Without a clear goal, the same social contract of liberties can lead to very different outcomes. And the group agreeing to that contract is a closed one, which is not enough for a theory of global justice.

Sen's move: it isn't necessary to agree on an ideal society at all — it is enough to be able to ask whether process X gets us closer to it than process Y.

Freedoms that build on freedoms

Development, for Sen, is the capacity to realise freedoms — where one kind of freedom expands another. Instrumental freedoms (political freedoms, economic facilities, social opportunities, transparency guarantees, protective security) build constitutive freedoms (avoiding starvation and undernourishment, being literate and numerate).

The end state is freedom itself, realised through other freedoms — not freedom as a route to equality of material goods alone.

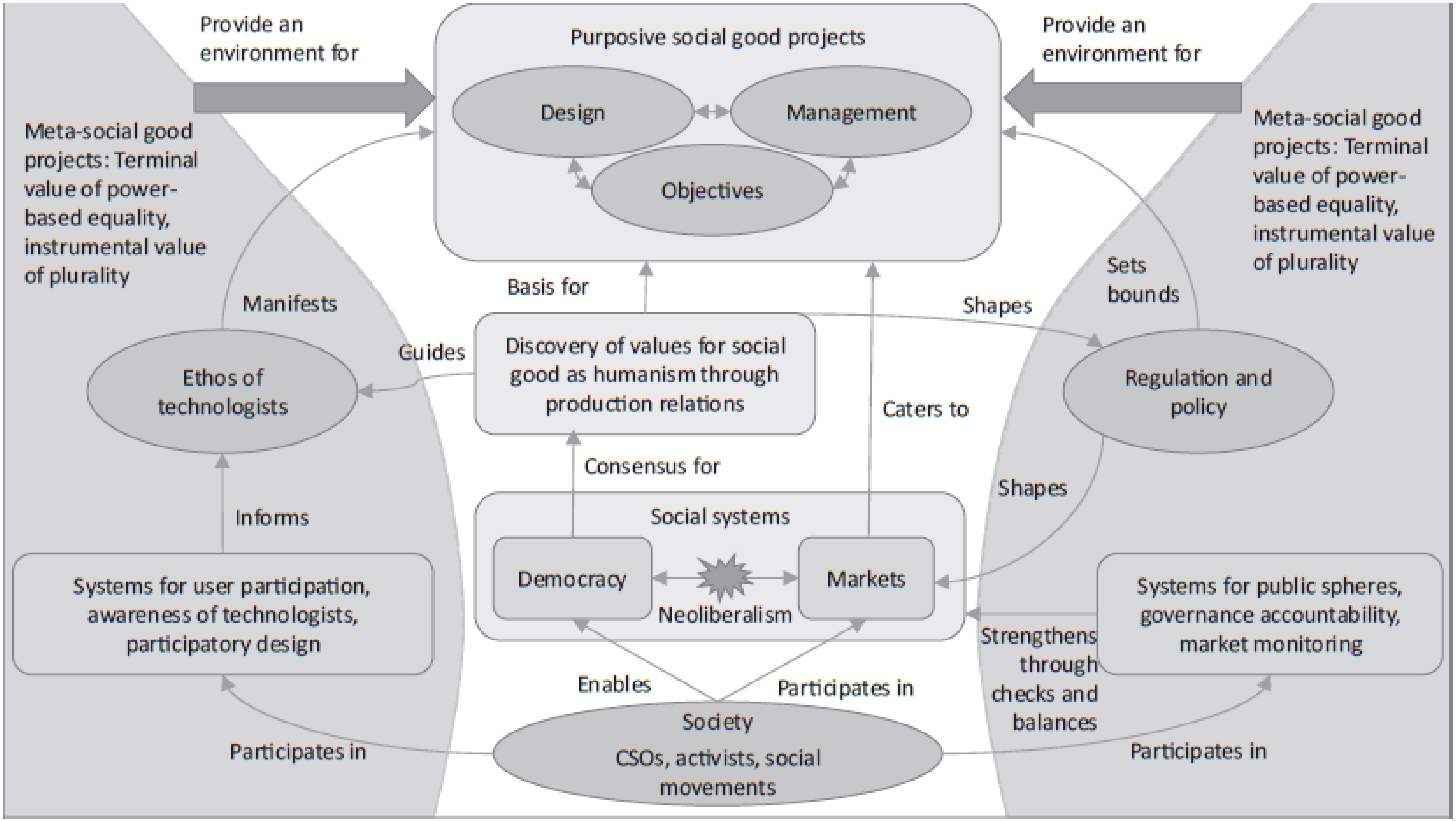
For ICTD projects: which instrumental freedom is this technology meant to expand — and which constitutive freedom does that, in turn, make possible?

What compromises social good — and what restores it

A Call to Technologists, 2022 — democracy as the throughline across every lens in this deck

1	**Society legitimises its systems**	Society legitimises the social systems within which it operates. These systems elect the dominant values that get reflected in the technologies society builds and uses — in their objectives, their design, and how they are managed.
2	**Structural injustice gets reproduced**	That election of values can reproduce structural injustice — an unregulated market is one example. Actors within market-driven systems amplify this further through accumulation, extending it into compromised institutions and a colonisation of society and culture that erodes its own values.
3	**Democracy and civil society restore it**	Social movements, labour collectives, and civil society are the actions that restore humanist values back into society. This is Young's point in another form: democracy is the only setup that allows the reflection and change needed — and the thread running through Ostrom's self-governing commons, PD's union roots, and Freire's dialogue.

Democracy is not one value alongside the others in this deck — it is the mechanism that keeps all the others accountable over time.



Provide an environment for
Purposive social good projects
Design
Management
Objectives
Provide an environment for
Meta-social good projects: Terminal value of power-based equality, instrumental value of plurality
Meta-social good projects: Terminal value of power-based equality, instrumental value of plurality
Basis for
Manifests
Sets bounds
Shapes
Guides
Discovery of values for social good as humanism through production relations
Ethos of technologists
Regulation and policy
Caters to
Consensus for
Shapes
Informs
Social systems
Democracy
Markets
Neoliberalism
Systems for user participation, awareness of technologists, participatory design
Systems for public spheres, governance accountability, market monitoring
Strengthens through checks and balances
Enables
Participates in
Society
CSOs, activists, social movements
Participates in
Participates in

The world we are designing toward

Seth, 2022 — Technology and (Dis)Empowerment: A Call to Technologists

Ownership & control	A digital commons or democratic coalition of local institutions steers the technology — not a large company. This extends Ostrom's design principles for governing a shared resource, and Beck's question of who shapes the agenda, to the technology itself.
Capability to operate	Users hold the skills to build, adapt, and manage the technology. This is Schumacher's intermediate technology and Marsden's HAPs, turned into a standing institutional capacity rather than a one-off transfer.
Power-based equality	A core terminal value: technologies are steered towards bringing power-based equality, with plurality as an instrumental value alongside it. This is Unwin's imperative — build for the power-weak, or the gap only widens — made explicit as a design goal.
Participatory practice	Design and management follow participatory practices, with constructive debate over specific social-good goals. This is the Scandinavian commitment to democracy and visible conflict, and Beck's insistence that participation itself be politically motivated.
A stronger public sphere	The experience is used to engage more widely with society — building public spheres between technologists and users, and drawing value from the market back to the commons. This is Freire's dialogue and praxis, scaled from one project to a democratic movement.

Discussion prompts for your own project proposal

1. In the application you are designing, what governance or ownership model decides who benefits — and who is accountable when it fails?
2. What would it mean for the community, not just your team, to be able to manage this technology after you leave?
3. Which stakeholders are marginalised in this space, and does your design foster their rights and voice — or work around them?
4. How will the problem itself be framed collaboratively, rather than arriving already defined by your team?

References

- Ostrom, E. (1990). Governing the Commons: The Evolution of Institutions for Collective Action. Cambridge University Press.
- Schumacher, E. F. (1973). Small Is Beautiful: Economics as if People Mattered. Blond & Briggs.
- Unwin, T. (2019). Why the notion of a Fourth Industrial Revolution is so problematic. Tim Unwin's Blog. https://timunwin.blog/2019/03/09/why-the-notion-of-a-fourth-industrial-revolution-is-so-problematic/
- Freire, P. (1970/1968). Pedagogy of the Oppressed. Herder and Herder.
- Guevara, E. — excerpt on communion with the peasantry of the Sierra Maestra, as cited in Freire, Pedagogy of the Oppressed.
- Seth, A. (2024-25). COL707: ICTs for Development and Sustainability — course slides “The role of the political economy of technology” and “What ethics to choose: Rights, power, equality, aspiration.” IIT Delhi.

References (continued)

- Gregory, J. (2003). Scandinavian Approaches to Participatory Design. International Journal of Engineering Education, 19(1), 62–74.
- Beck, E. (2002). P for Political — Participation Is Not Enough. Scandinavian Journal of Information Systems, 14(1).
- Hayes, G. R. (2011). The Relationship of Action Research to Human-Computer Interaction. ACM Transactions on Computer-Human Interaction, 18(3).
- Seth, A. (2022). Technology and (Dis)Empowerment: A Call to Technologists. Emerald Publishing.

References (continued)

- Rawls, J. (1971). A Theory of Justice. Harvard University Press.
- Young, I. M. (1990). Justice and the Politics of Difference. Princeton University Press.
- Sen, A. (1999). Development as Freedom. Oxford University Press.